GRUB ISSUE

There are two different GRUB shells. Depending on the root cause of the problem, you might find yourself at a GRUB shell prompt:

grub>

or at a GRUB rescue shell prompt:

grub rescue>

The following troubleshooting steps are related to the grub shell prompt:

The GRUB shell is a command-line interface that allows you to use a limited set of Linux commands. If you find yourself here instead of your usual login screen, it indicates that there is an issue with the boot process.

Before going to load the vmlinuz, one thing we need to clarify is whether the system is a manual partition or an LVM partition, because the kernel needs to load into the correct partition otherwise, it will not boot.

If the device name is different, it may be:

- /dev/vda2 (if KVM/cloud)
- /dev/sda2 (normal server)

A very important thing while doing manually booting:

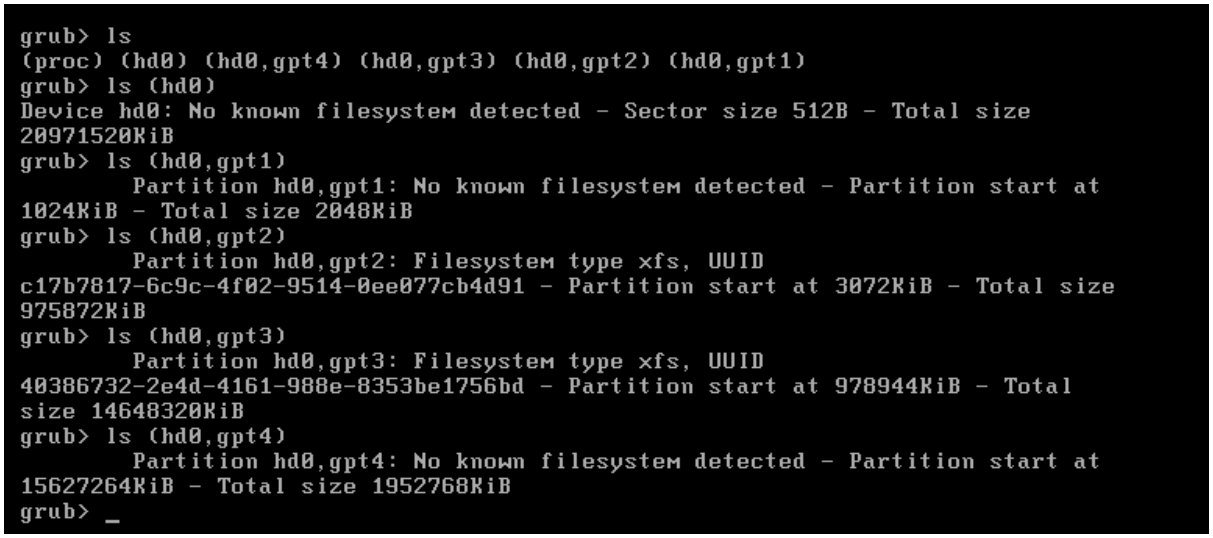
Never assume /dev/sda. Always match actual disk type:

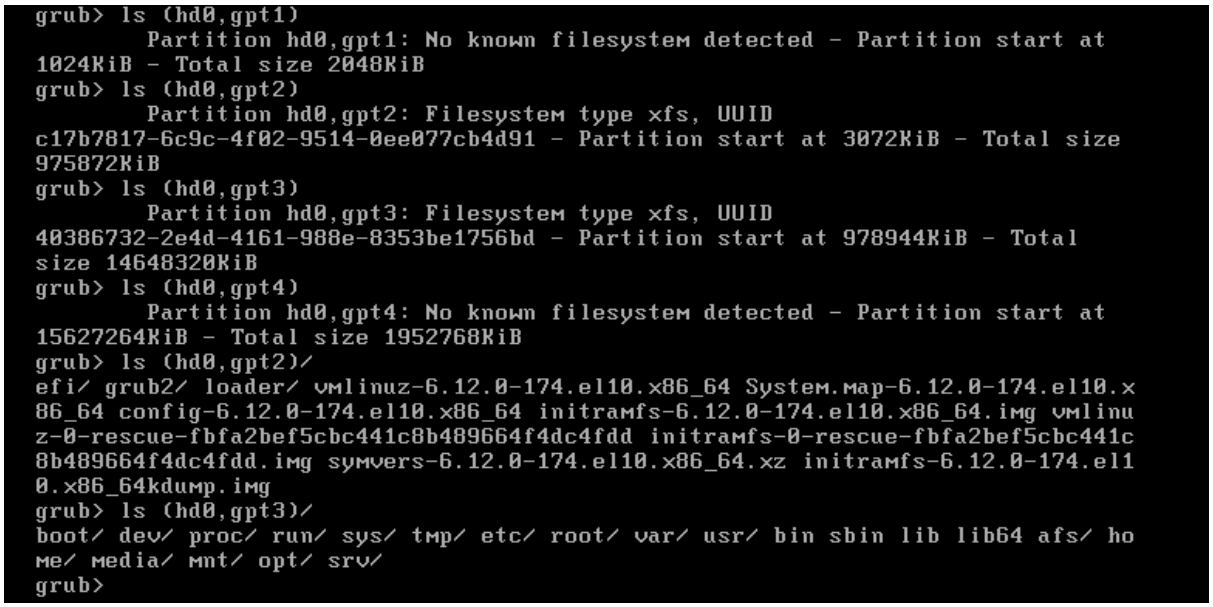
- SATA → sda
- VirtIO → vda
- NVMe → nvme0n1pX

In my case, it's an **"sda".**

If it is a Manual partition, follow the steps below:
At the grub screen:







From the GRUB view, the disk is (hd0). Usually, that maps to /dev/sda.

If It Fails With “cannot find root device”. Then that system is using LVM.

(hd0) is the first hard drive.

(hd0,gpt1) → small 2MB → likely BIOS boot partition

(hd0,gpt2) → XFS → contains /boot, vmlinuz, initramfs

(hd0,gpt3) → XFS → contains /boot /dev /proc /run /sys /tmp /etc /root /var /usr ... → This is root filesystem

(hd0,gpt4) → no filesystem → possibly swap

From the output:

(hd0,gpt3) → contains /etc /usr /var

This is the root (/) partition.

(hd0,gpt2) → contains vmlinuz + initramfs

Partition	Purpose
gpt1	BIOS boot
gpt2	/boot
gpt3	/
gpt4	swap

Set root to /boot partition:

1. set root=(hd0,gpt2)

Load kernel type (press TAB to autocomplete):

2. linux /vmlinuz-6.12.0-174.el10.x86_64 root=/dev/sda3 ro

Important:
The root partition is gpt3 → so /dev/sda3

Load initramfs:

3. initrd /initramfs-6.12.0-174.el10.x86_64.img

Boot:

4. boot

If it Boots Successfully. Then inside the OS:

5. sudo grub2-install /dev/sda

6. sudo grub2-mkconfig -o /boot/grub2/grub.cfg

```
PS C:\Users\W2556> ssh root@192.168.163.129
root@192.168.163.129's password:
Web console: https://localhost:9090/ or https://192.168.163.129:9090/

Last login: Fri Feb 13 16:48:06 2026 from 192.168.163.1
root@localhost:~# lsblk
NAME   MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda      8:0    0   20G  0 disk
├─sda1   8:1    0    2M  0 part
├─sda2   8:2    0  953M  0 part /boot
├─sda3   8:3    0   14G  0 part /
└─sda4   8:4    0  1.9G  0 part [SWAP]
sr0     11:0    1  8.6G  0 rom
root@localhost:~# cd /boot/
root@localhost:/boot# ls
config-6.12.0-174.el10.x86_64  initramfs-0-rescue-fbfa2bef5cbc441c8b489664f4dc4fdd.img  loader                                vmlinuz-0-rescue-fbfa2bef5cbc441c8b489664f4dc4fdd
efi                            initramfs-6.12.0-174.el10.x86_64.img                    symvers-6.12.0-174.el10.x86_64.xz     vmlinuz-6.12.0-174.el10.x86_64
grub2                          initramfs-6.12.0-174.el10.x86_64kdump.img               System.map-6.12.0-174.el10.x86_64
root@localhost:/boot# cd grub2/
root@localhost:/boot/grub2# ll
total 24
-rw-r--r--. 1 root root   64 Feb 12 19:47 device.map
drwxr-xr-x. 2 root root   25 Feb 12 19:47 fonts
-rw-r--r--. 1 root root 1024 Feb 12 19:47 grubenv
drwxr-xr-x. 2 root root 8192 Feb 12 19:47 i386-pc
drwxr-xr-x. 2 root root 4096 Feb 12 19:47 locale
root@localhost:/boot/grub2# grub2-mkconfig -o /boot/grub2/grub.cfg
Generating grub configuration file ...
Adding boot menu entry for UEFI Firmware Settings ...
done
root@localhost:/boot/grub2# ll
total 32
-rw-r--r--. 1 root root   64 Feb 12 19:47 device.map
drwxr-xr-x. 2 root root   25 Feb 12 19:47 fonts
-rw-------. 1 root root 7366 Feb 13 16:53 grub.cfg
-rw-r--r--. 1 root root 1024 Feb 12 19:47 grubenv
drwxr-xr-x. 2 root root 8192 Feb 12 19:47 i386-pc
drwxr-xr-x. 2 root root 4096 Feb 12 19:47 locale
root@localhost:/boot/grub2# grub2-install /dev/sda
Installing for i386-pc platform.
Installation finished. No error reported.
root@localhost:/boot/grub2#
```

For UEFI systems:

7. sudo grub2-mkconfig -o /boot/efi/EFI/*/grub.cfg

OR

If it is an LVM partition, follow the steps below:

```
                              GRUB version 2.12

   Minimal BASH-like line editing is supported. For the first word, TAB
   lists possible command completions. Anywhere else TAB lists possible
   device or file completions. To enable less(1)-like paging, "set
   pager=1".


grub>
```

```
grub> ls
(proc) (hd0) (hd0,gpt3) (hd0,gpt2) (hd0,gpt1)
grub> ls (hd0)
Device hd0: No known filesystem detected - Sector size 512B - Total size
20971520KiB
grub> ls (hd0,gpt1)
        Partition hd0,gpt1: No known filesystem detected - Partition start at
1024KiB - Total size 1024KiB
grub> ls (hd0,gpt2)
        Partition hd0,gpt2: Filesystem type xfs, UUID
f932f580-c623-4c3b-a4f1-745bb1381a75 - Partition start at 2048KiB - Total size
1048576KiB
grub> ls (hd0,gpt3)
        Partition hd0,gpt3: No known filesystem detected - Partition start at
1050624KiB - Total size 19919872KiB
grub>
```

Hard Disk Details:

- (hd0,gpt1) → No known filesystem (probably EFI partition or BIOS boot)
- (hd0,gpt2) → Filesystem type xfs, UUID shown
- (hd0,gpt3) → No known filesystem (maybe swap or LVM physical volume)

That **actual Linux root partition is located** in **(hd0,gpt2)**.

Now recover it manually:

```
grub> ls
(proc) (hd0) (hd0,gpt3) (hd0,gpt2) (hd0,gpt1)
grub> ls (hd0)
Device hd0: No known filesystem detected - Sector size 512B - Total size
20971520KiB
grub> ls (hd0,gpt1)
        Partition hd0,gpt1: No known filesystem detected - Partition start at
1024KiB - Total size 1024KiB
grub> ls (hd0,gpt2)
        Partition hd0,gpt2: Filesystem type xfs, UUID
f932f580-c623-4c3b-a4f1-745bb1381a75 - Partition start at 2048KiB - Total size
1048576KiB
grub> ls (hd0,gpt3)
        Partition hd0,gpt3: No known filesystem detected - Partition start at
1050624KiB - Total size 19919872KiB
grub> ls (hd0,gpt2)/
efi/ grub2/ loader/ vmlinuz-6.12.0-174.el10.x86_64 System.map-6.12.0-174.el10.x
86_64 config-6.12.0-174.el10.x86_64 initramfs-6.12.0-174.el10.x86_64.img vmlinu
z-0-rescue-710154e6eba54aa89649a67f9a96ab67 initramfs-0-rescue-710154e6eba54aa8
9649a67f9a96ab67.img symvers-6.12.0-174.el10.x86_64.xz initramfs-6.12.0-174.el1
0.x86_64kdump.img
grub>
```

Here in (hd0,gpt2)/ it displays the EFI, GRUB2, vmlinuz, initramfs, and more. This is the /boot directory, as it contains all boot-related files.

Now GRUB wants to know where the root is:

8. set root=(hd0,gpt2)

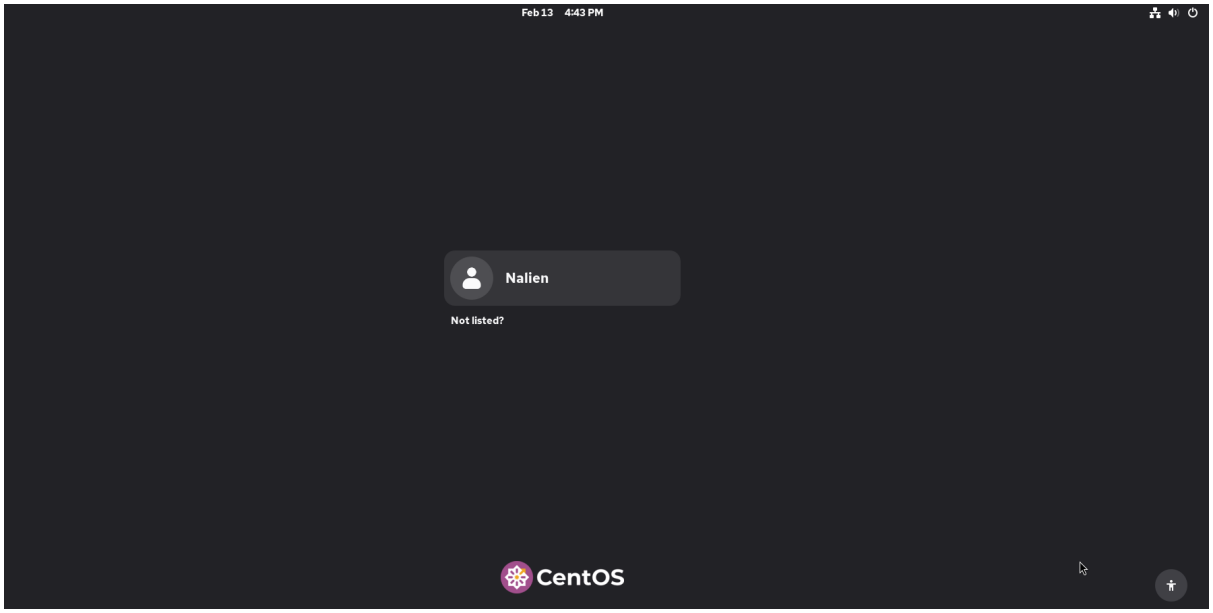
In my case the root filesystem is probably on **gpt3**, not gpt2.

Because:

- gpt2 = boot partition (small, ~1GB)
- gpt3 = large partition (likely root LVM or root FS)

At the grub screen:

9. set root=(hd0,gpt2)
10. linux (hd0,gpt2)/vmlinuz-6.12.0-174.el10.x86_64 root= /dev/mapper/<VGNAME>-root ro
11. initrd (hd0,gpt2)/initramfs-6.12.0-174.el10.x86_64.img
12. boot



After the system successfully Boots run the following commands:

We must fix GRUB permanently so you don't land in this situation again.

13. grub2-mkconfig -o /boot/grub2/grub.cfg

If a UEFI system:

14. grub2-mkconfig -o /boot/efi/EFI/*/grub.cfg

In my case, it's an **"**nvme0n1**".**

Then reinstall the bootloader:

15. grub2-install /dev/nvme0n1

```
root@localhost:/boot/grub2# grub2-mkconfig -o /boot/grub2/grub.cfg
Generating grub configuration file ...
Adding boot menu entry for UEFI Firmware Settings ...
done
root@localhost:/boot/grub2# ll
total 44
-rw-r--r--. 1 root root    68 Jan  7 14:48 device.map
drwxr-xr-x. 2 root root    25 Jan  7 14:48 fonts
-rw-------. 1 root root  7242 Feb 12 18:39 grub.cfg
-rw-------. 1 root root  1024 Feb  9 15:57 grubenv
drwxr-xr-x. 2 root root 20480 Feb 12 15:29 i386-pc
drwxr-xr-x. 2 root root  4096 Feb 12 15:29 locale
root@localhost:/boot/grub2# grub2-install /dev/nvme0n1
Installing for i386-pc platform.
Installation finished. No error reported.
```

The following troubleshooting steps are related to the grub rescue shell prompt:



This is a GRUB corruption / broken bootloader problem.

The key error is:

file '/grub2/i386-pc/normal.mod' not found

Entering rescue mode...

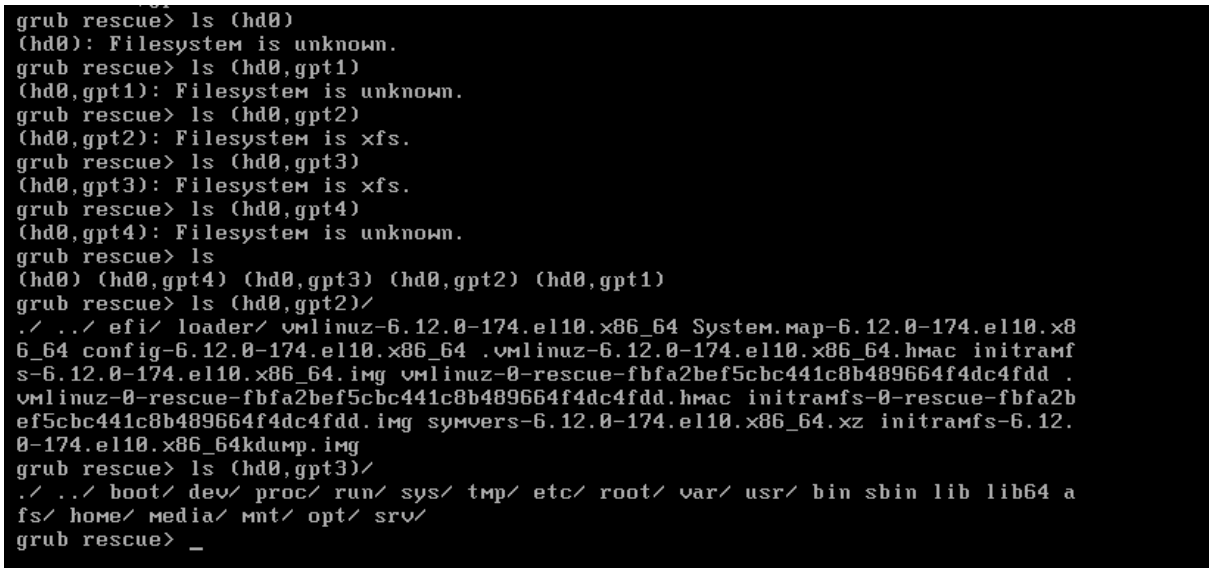
grub rescue>

That means:

GRUB is installed, but it cannot find its modules. Either /boot is corrupted, or a wrong disk/partition is being loaded, or GRUB was partially overwritten.

This is not a simple issue, but it's fixable.

Type the ls command and find how many hard disks and partitions are there.



(hd0,gpt2) → contains /efi, /loader, kernel images → this is /boot partition

(hd0,gpt3) → contains /boot /dev /etc /usr ... → this is root (/) partition

Set Correct Root and Prefix:

If doesn't know, just type the "set" command in the terminal output will show the correct root and prefix details. In my case

(hd0,gpt2) → contains vmlinuz and initramfs

So do this carefully:

- set root=(hd0,gpt2)
- set prefix=(hd0,gpt2)/grub2
- insmod normal
- normal

If insmod normal works → it will bring back the normal GRUB menu.

If insmod normal Fails proceed with the following steps to resolve the issue:

```
grub rescue> set root=(hd0,gpt2)
grub rescue> set prefix=(hd0,gpt2)/grub2
grub rescue> insmod normal
error: ../../grub-core/fs/fshelp.c:257:file '/grub2/i386-pc/normal.mod' not foun
d.
```

The error:

file '/grub2/i386-pc/normal.mod' not found

This tells us something very important:

The GRUB is BIOS (i386-pc version). The grub modules are missing or corrupted. This is not a small config issue. The GRUB core image cannot find its modules. Manual loading will NOT work anymore.

What Actually Happened:

The GRUB core (installed in MBR/BIOS area) is looking for: /boot/grub2/i386-pc/normal.mod

But either:

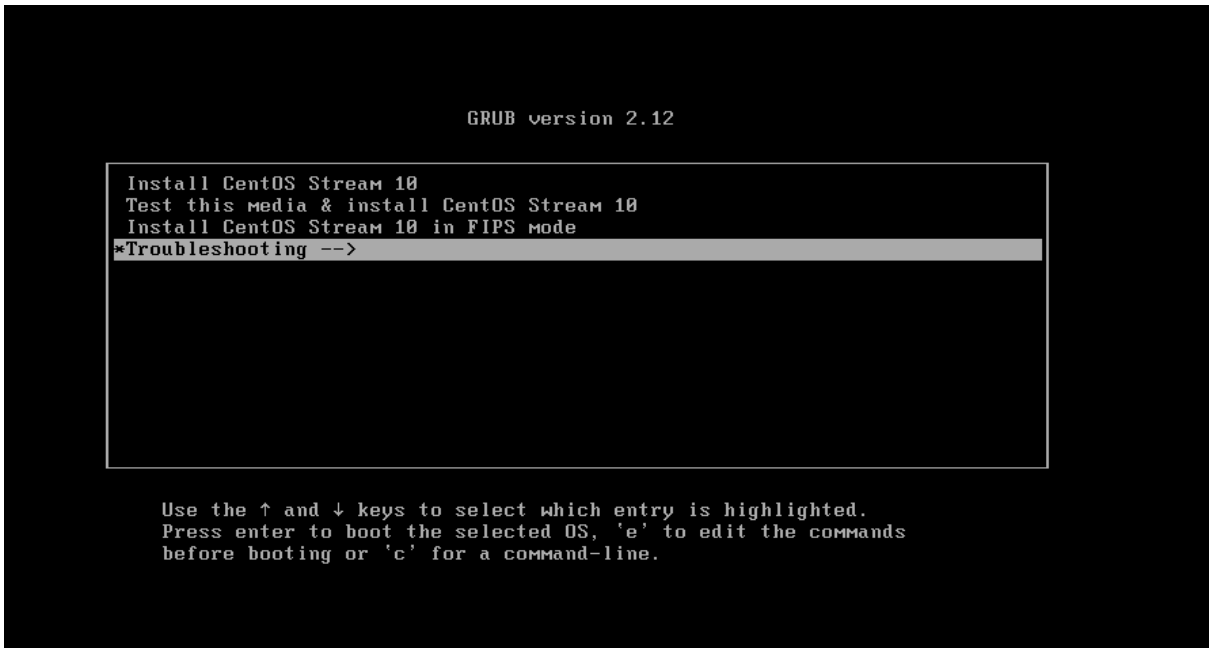
/boot/grub2/i386-pc/ is missing or modules are damaged or grub2 package partially removed or disk corruption happened

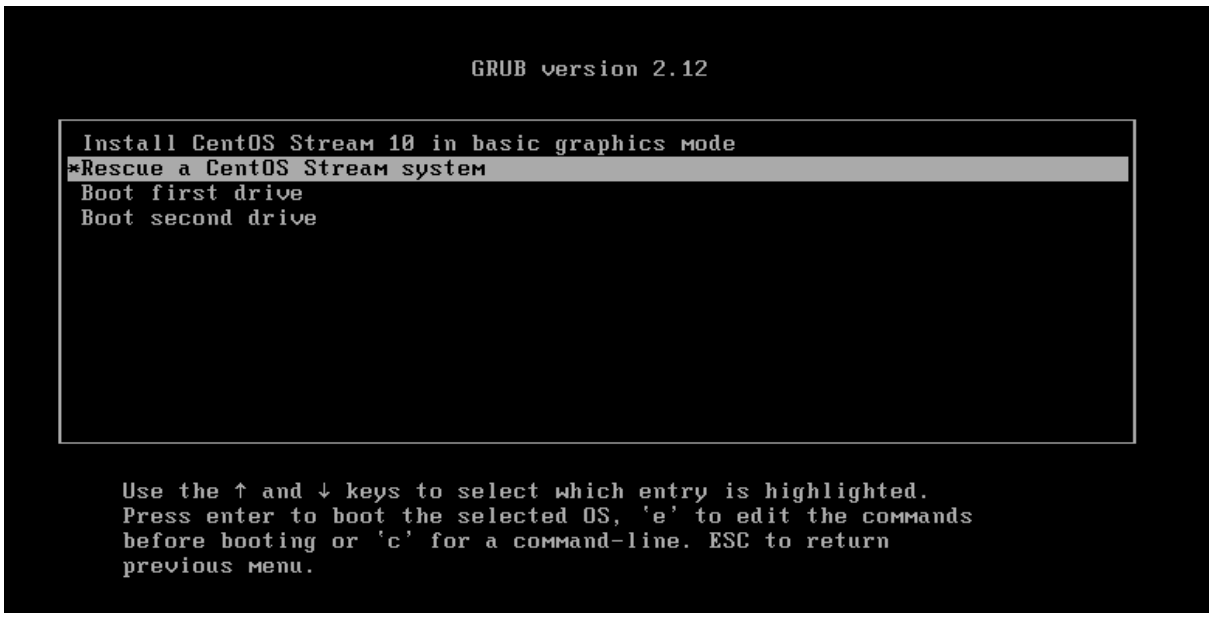
Since (hd0,gpt2) clearly contains /boot, but GRUB cannot load modules → reinstall is required.

Step 1 — Boot from CentOS ISO

Choose:

Troubleshooting → Rescue a CentOS system





Step 2 — Enter Chroot

After it mounts your system:

- chroot /mnt/sysimage

Now inside your installed system.

```
Please press ENTER to get a shell:
bash-5.2# lsblk
NAME   MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
loop0    7:0    0 746.3M  1 loop /run/rootfsbase
sda      8:0    0    20G  0 disk
├─sda1   8:1    0     2M  0 part
├─sda2   8:2    0   953M  0 part /mnt/sysroot/boot
│                               /mnt/sysimage/boot
├─sda3   8:3    0    14G  0 part /mnt/sysroot
│                               /mnt/sysimage
└─sda4   8:4    0   1.9G  0 part
sr0     11:0    1   8.6G  0 rom  /run/install/repo
bash-5.2# chroot /mnt/sysimage
bash-5.2# lsblk
NAME   MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
loop0    7:0    0 746.3M  1 loop
sda      8:0    0    20G  0 disk
├─sda1   8:1    0     2M  0 part
├─sda2   8:2    0   953M  0 part /boot
├─sda3   8:3    0    14G  0 part /
└─sda4   8:4    0   1.9G  0 part
sr0     11:0    1   8.6G  0 rom
bash-5.2# dnf reinstall grub2 grub2-tools -y_
[anaconda]1:main* 2:shell  3:log  4:storage-log  5:program-log  6:packaging-log
```

```
Starting installer, one moment...
libreport is not available in this environment - bug reporting disabled
anaconda 40.22.3.39-1.el10 for CentOS Stream 10 started.
 * installation log files are stored in /tmp during the installation
 * shell is available on TTY2 and in second TMUX pane (ctrl+b, then press 2)
 * when reporting a bug add logs from /tmp as separate text/plain attachments
================================================================================
================================================================================
Rescue

The rescue environment will now attempt to find your Linux installation and
mount it under the directory : /mnt/sysroot.  You can then make any changes
required to your system.  Choose '1' to proceed with this step.
You can choose to mount your file systems read-only instead of read-write by
choosing '2'.
If for some reason this process does not work choose '3' to skip directly to a
shell.



1) Continue
2) Read-only mount
3) Skip to shell
4) Quit (Reboot)

Please make a selection from the above: 1
================================================================================
================================================================================
Rescue Shell

Your system has been mounted under /mnt/sysroot.

If you would like to make the root of your system the root of the active system,
run the command:

        chroot /mnt/sysroot

Warning: The rescue shell will trigger SELinux autorelabel on the subsequent
boot. Add "enforcing=0" on the kernel command line for autorelabel to work
properly.
When finished, please exit from the shell and your system will reboot.

Please press ENTER to get a shell:
bash-5.2# _

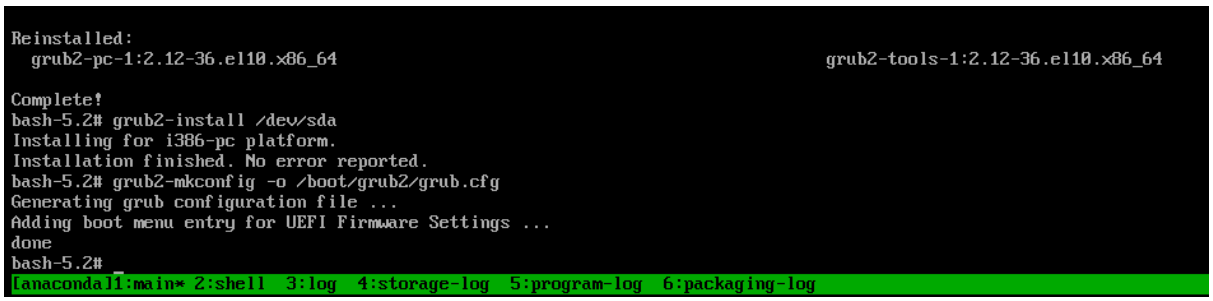
[anaconda]1:main* 2:shell  3:log  4:storage-log  5:program-log  6:packaging-log                 Switch tab: Alt+Tab | Help: F1
```

Step 3 — Reinstall GRUB (BIOS Mode)

Since the error shows i386-pc, this is BIOS mode.

Run:

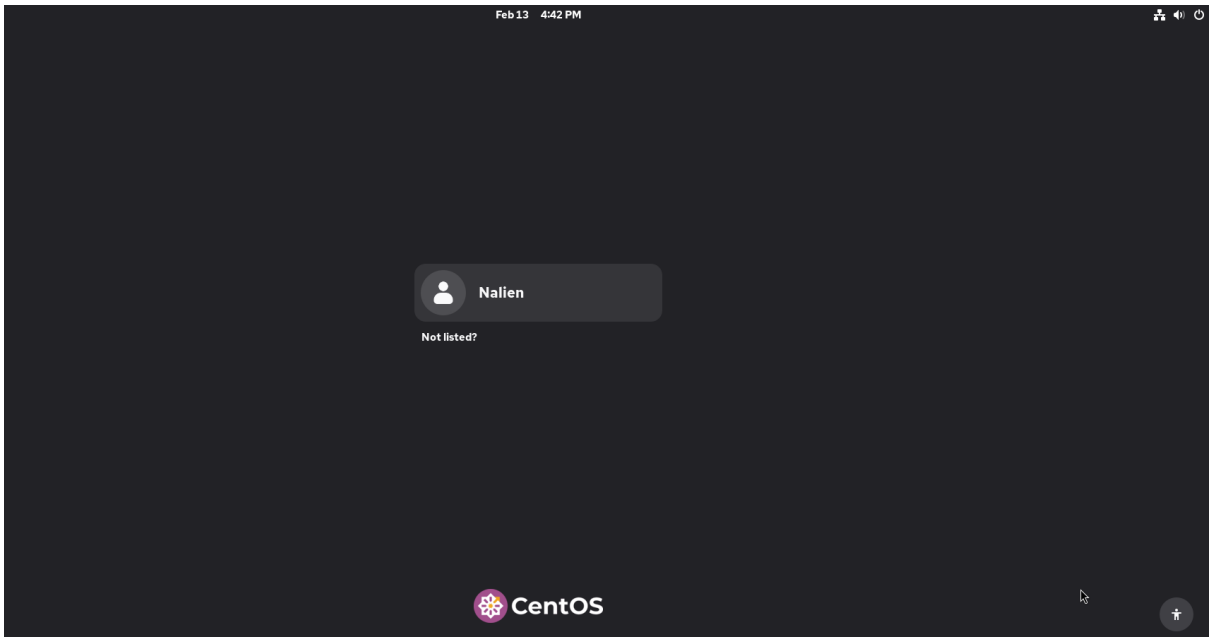
- dnf reinstall grub2 grub2-tools -y
- grub2-install /dev/sda
- grub2-mkconfig -o /boot/grub2/grub.cfg



Exit and Reboot

- exit
- reboot

Remove ISO.



How to find if it's a sda. vda or nvme in grub screen?

When in GRUB, Linux device names like /dev/sda, /dev/vda, /dev/nvme0n1 do NOT exist yet. GRUB has its own naming system:

(hd0)

(hd0,gpt1)

(hd0,gpt2)

So cannot directly “see sda or nvme” inside GRUB.

But can determine it logically.

Method 1 — Check from Kernel Boot Messages (Most Reliable):

After loading the kernel and it starts printing logs, look carefully at the disk detection lines.

Examples:

If SATA / SCSI

You will see:

sd 0:0:0:0: [sda] ...

→ That means /dev/sda

If VirtIO (KVM)

You will see:

vda: vda1 vda2

→ That means /dev/vda

If NVMe

You will see:

nvme0n1: p1 p2 p3

Method 2 — Use GRUB UUID

Inside GRUB, you saw:

Filesystem type xfs, UUID f932f580-....

Later in the emergency shell:

- ls /dev/disk/by-uuid/

So, you map UUID → device name.

Method 3 — Use Rescue Mode

Boot from ISO → rescue mode.

Then run:

- lsblk -f

That shows real device names cleanly.

Important Concept

GRUB disk names:

GRUB	LINUX NAME
(hd0)	First disk
(hd0,gpt1)	First disk p1
(hd0,gpt2)	First disk p2

But whether that disk becomes: /dev/sda, /dev/vda, /dev/nvme0n1

Depends on the hardware type, the hypervisor, Driver loaded by the kernel

GRUB does not care about Linux naming.

Best Practice When Manually Booting:

Instead of guessing use UUID (safest method):

Inside GRUB:

- linux ... root=UUID=<your-root-uuid> ro

Example:

- linux (hd0,gpt2)/vmlinuz-... root=UUID=xxxx-xxxx ro

Cannot directly see sda/vda/nvme inside GRUB.

1. Watching kernel boot logs
2. Checking UUID mapping
3. Using rescue environment